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Multi-Platform Databases in the Cloud: How Workloads Impact Decisions

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I'm the AI and Multi-Platform Advocate at Redgate Software. Along with presenting on open source, AI and Oracle products, I also work with the engineering and product teams on Redgate solutions.

I've been in the tech industry for over 27 years and am known online under my handle, DBAkevlar.



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I'm an innovative Principal Cloud Solution Architect with extensive experience in designing and implementing end-to-end technology projects across multiple industries.

I focus on success in leading cross-functional teams, fostering collaboration to ensure seamless execution of complex cloud architectures. Community-driven and dynamic presenter, with extensive international footprint, recognized for delivering insightful perspectives to diverse audiences in 30+ countries.



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Agenda: Key Topics Overview

- Understanding Multi-Platform Cloud Database Choices
- Mapping Workloads to Cloud Service Models
- Critical Considerations and Common Pitfalls in Cloud Migrations
- Optimizing Costs and Planning Before Migration
- Advanced Migration Tools and the Role of AI in Cloud Transitions

UNDERSTANDING MULTI- PLATFORM CLOUD DATABASE CHOICES

Why Multi-Platform Databases Matter

Enterprises rarely standardize on one DB

Cloud adoption accelerates polyglot persistence

Cost, compliance, performance vary by workload

Skills and tooling drive decisions

OVERVIEW OF SAAS, PAAS, AND IAAS DATABASE MODELS

SaaS Databases

SaaS provides fully managed databases requiring minimal user intervention, ideal for simple and straightforward workloads.

PaaS Database Platforms

PaaS offers a managed platform with more customization options, suited for workloads of moderate complexity.

IaaS Infrastructure

IaaS delivers raw infrastructure requiring full user management, ideal for complex workloads with specific needs.



Cloud Service Examples

Category	Database	Cloud Vendor
SaaS	CosmosDB	Azure
SaaS	Aurora Serverless	AWS
SaaS	Firestore	AWS
PaaS	Azure SQL DB	Azure
PaaS	Managed Instance	Azure
PaaS	Cloud SQL	Google Cloud
IaaS	Oracle on VM	Azure
PaaS	Oracle RDS	AWS
IaaS	Postgres EC2	AWS
IaaS	MySQL on GCP	Google Cloud
PaaS	Azure DB for PostgreSQL	Azure

Decision Drivers for Cloud Vendors



Azure: Hybrid, enterprise integrations



AWS: Broadest DB catalog



Google: Analytics & AI/ML integration



Drivers: Cost, compliance, performance, lock-in



Case Scenarios

Startup SaaS → Firestore (SaaS)

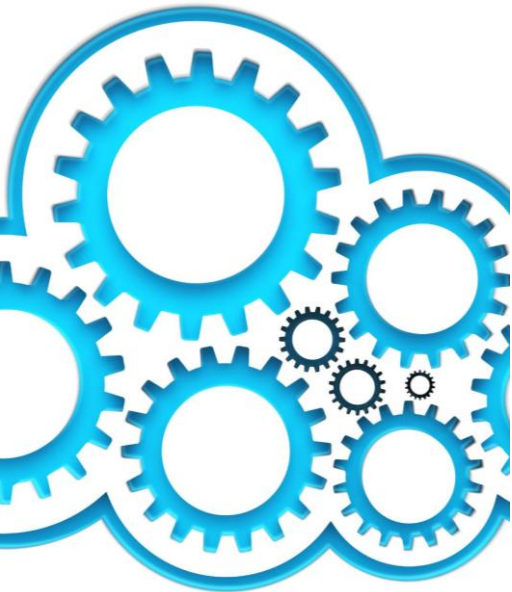
Retail Ecommerce → AWS RDS
(PaaS, refactor)

Financial Trading → Oracle
Exadata Cloud (IaaS/infra)

Workload Spectrum

- Small workloads: simple apps, predictable usage
- Medium workloads: higher concurrency, scaling challenges
- Large workloads: mission-critical, specialized infra needs

KEY FACTORS INFLUENCING MIGRATION DECISIONS



Workload Complexity

Workload complexity affects migration strategy and cloud service model selection decisions.

Performance Needs

Performance requirements determine the suitable cloud resources and service tiers for workloads.

Security Requirements

Security needs influence migration decisions ensuring data protection and compliance in the cloud.

Operational Capabilities

Operational capabilities guide the management and support approach for migrated workloads.

SaaS for Small Workloads

- Minimal customization
- Low operational overhead
- Scaling abstracted
- Examples: Analytics dashboards, lightweight web apps

PaaS for Growing/Demanding Workloads

- Handles higher demand with tuning flexibility
- Requires refactoring for compatibility
- Balance of simplicity and control
- Examples: OLTP, ecommerce

IaaS for Specialized or Large Workloads

- Full VM/database control
- Supports specialized features (RAC, configs)
- Requires significant rewrite for cloud
- Examples: Core banking, ERP, real-time analytics

Workload Classification: Small, Mid-Size, and Complex Workloads



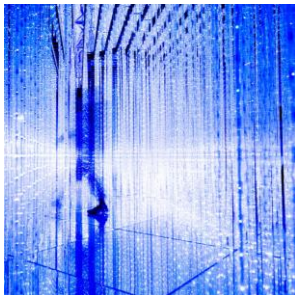
Small Workloads

Small workloads are simple and require low resources, suited for straightforward migration strategies.



Mid-Size Workloads

Mid-size workloads have moderate resource needs and complexity, requiring balanced migration and service models.



Complex Workloads

Complex workloads are large-scale with high customization, demanding advanced migration approaches and flexible service models.

MAPPING WORKLOADS TO CLOUD SERVICE MODELS

MIGRATING SMALL, SIMPLE WORKLOADS TO SAAS SOLUTIONS



Ease of Migration

SaaS solutions simplify the migration process for small workloads, reducing complexity and downtime.

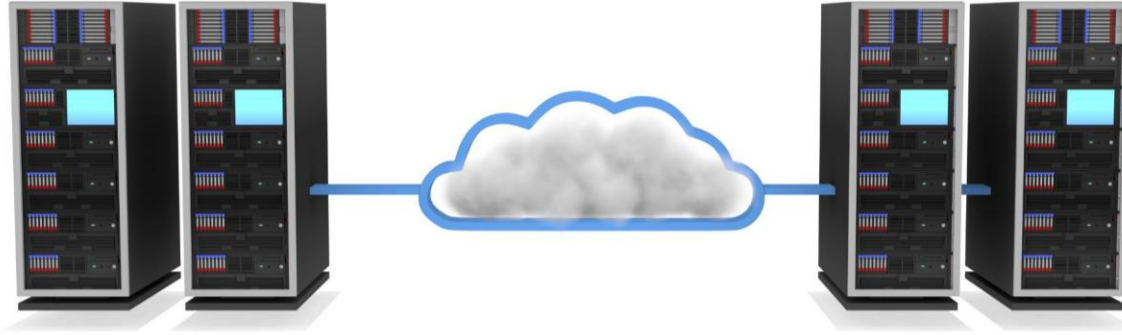
Minimal Management Required

SaaS databases require minimal maintenance, freeing up resources and reducing operational overhead.

Cost-Effectiveness

Using SaaS for small workloads reduces costs by eliminating hardware investments and optimizing resource usage.

HANDLING MID-SIZE WORKLOADS WITH MODERATE COMPLEXITY VIA PAAS



Control and Customization

Mid-size workloads benefit from PaaS due to enhanced control and the ability to customize solutions for specific needs.

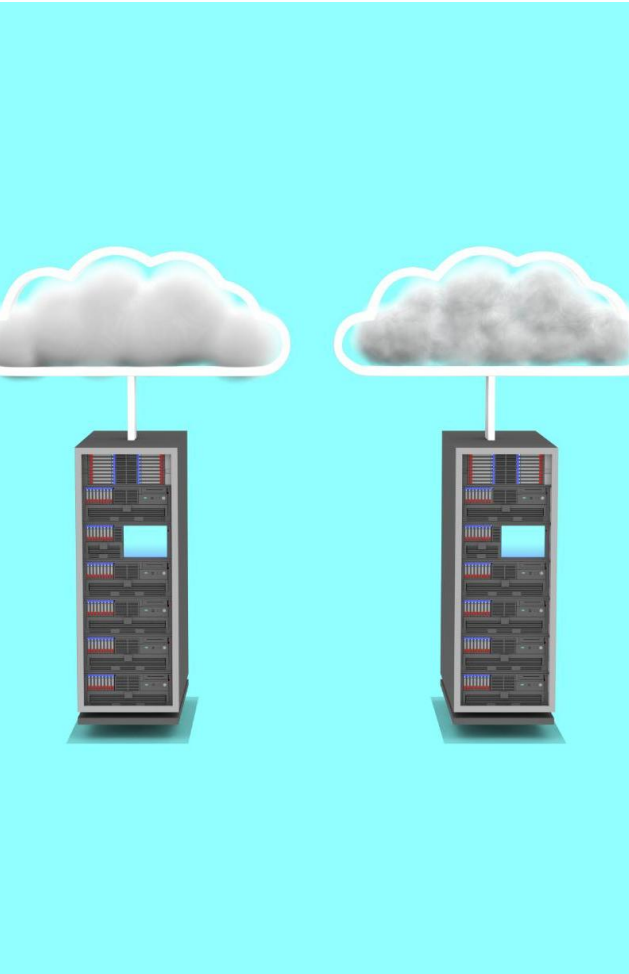
Tailoring PaaS Offerings

Tailoring PaaS solutions helps meet moderate workload demands while optimizing resource allocation and performance.

Operational Efficiency

Using PaaS for mid-size workloads maintains operational efficiency through streamlined processes and automation.

REFACTORING AND MIGRATING THE LARGEST, MOST COMPLEX WORKLOADS TO IAAS



Refactoring for Complexity

Refactoring simplifies complex workloads to optimize them for cloud infrastructure and improve manageability.

Full Infrastructure Control

IaaS provides complete control over infrastructure, enabling customization and fine-tuning of resources.

Managing Complexity

Effective strategies include modular design and automation to handle workload complexity during migration.

Maintaining Performance

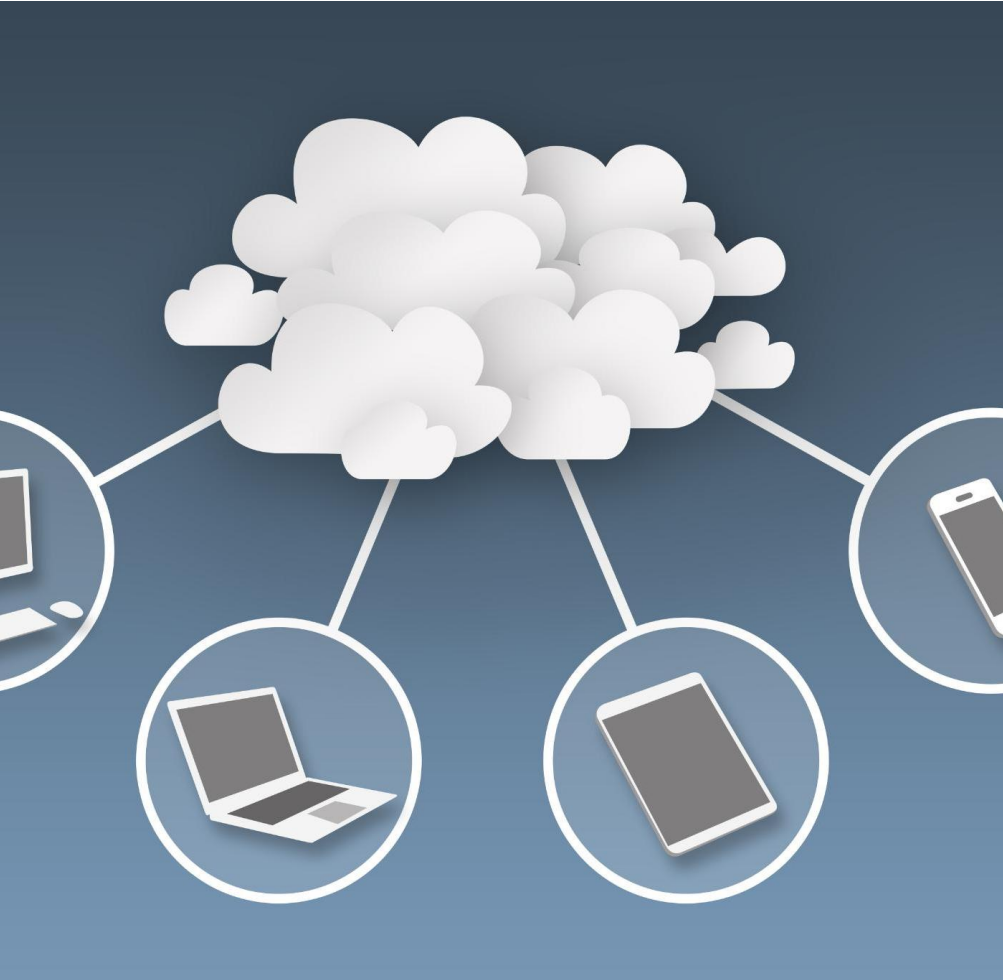
Performance is maintained by continuously monitoring and optimizing resource allocation in IaaS.

Best Practices

- Evaluate workload size & refactor effort
- Start SaaS → move up if needed
- Multi-cloud strategies
- Governance & skills alignment
- Use monitoring & CI/CD tools

CRITICAL CONSIDERATIONS AND COMMON PITFALLS IN CLOUD MIGRATIONS

SCALING LIMITS AND PERFORMANCE BOTTLENECKS IN DIFFERENT MODELS



Cloud Model Constraints

Different cloud service models impose unique scalability and performance limitations to consider.

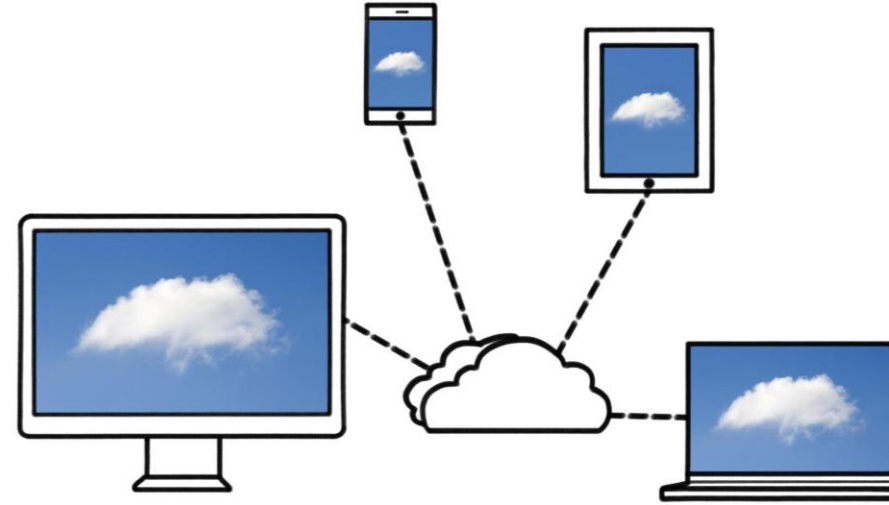
Identifying Bottlenecks

Recognizing performance bottlenecks early helps optimize resource allocation and service delivery.

Meeting Workload Demands

Choosing platforms that match workload requirements ensures efficiency and scalability.

BENEFITS OF SIMPLICITY AND REDUCING OPERATIONAL OVERHEAD



Improved Reliability

Simplified system architectures reduce failure points and enhance overall service reliability for users.

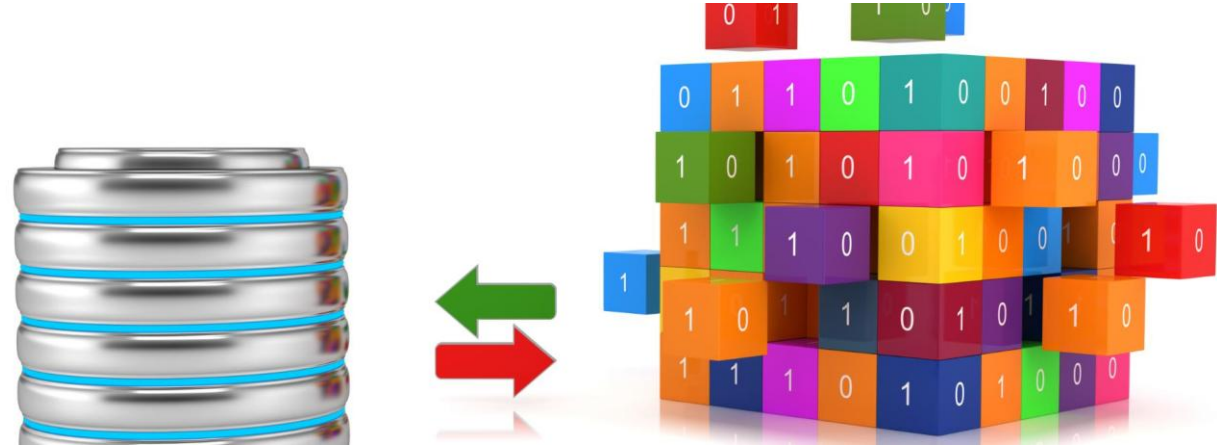
Lower Operational Costs

Minimizing manual management tasks decreases operational expenses and resource requirements.

Operational Efficiency

Utilizing appropriate cloud models improves efficiency by automating routine tasks and management.

FREQUENT MIGRATION CHALLENGES AND HOW TO AVOID THEM



Common Migration Issues

Data loss, downtime, and misaligned configurations are frequent problems faced during migrations.

Proactive Planning

Careful planning before migration helps minimize risks and ensures smoother transitions.

Testing and Tools

Thorough testing and selecting appropriate tools reduce errors and downtime during migration.

OPTIMIZING COSTS AND PLANNING BEFORE MIGRATION

IMPORTANCE OF COST OPTIMIZATION BEFORE MIGRATION



Identify Potential Savings

Analyzing costs before migration helps recognize areas where expenses can be reduced effectively.

Prevent Overspending

Pre-migration cost review helps avoid unexpected expenses and budget overruns during migration.

Balance Performance and Budget

Cost optimization ensures migration plans achieve good performance without exceeding budget limits.

ANALYZING AND RIGHT-SIZING WORKLOADS FOR CLOUD ENVIRONMENTS



Workload Resource Matching

Aligning workload requirements with appropriate cloud resources ensures efficient performance and scalability.

Cost Avoidance

Right-sizing prevents overprovisioning, reducing unnecessary cloud expenditure and optimizing budgets.

Operational Efficiency

Properly sized workloads improve cloud operation efficiency and reduce resource wastage.

TOOLS AND PRACTICES FOR ACCURATE COST FORECASTING



Cloud-Native and Third-Party Tools

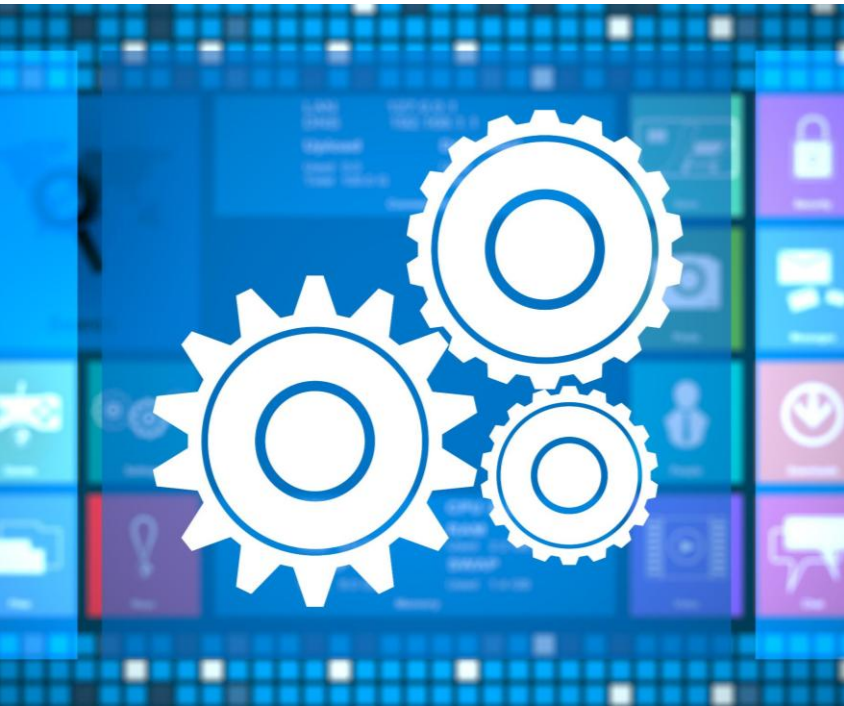
Using cloud-native and third-party tools enhances visibility into costs for more accurate forecasting.

Regular Monitoring and Adjustments

Continuous monitoring and timely adjustments ensure budget compliance after migration.

ADVANCED MIGRATION TOOLS AND THE ROLE OF AI IN CLOUD TRANSITIONS

OVERVIEW OF LEADING MIGRATION TOOLS AND AUTOMATION PLATFORMS



Automation for Assessment

Migration tools automate assessment tasks to efficiently analyze data sources and readiness.

Data Transfer Automation

Platforms streamline and secure data transfer processes to minimize downtime and data loss.

Validation and Error Reduction

Automation platforms validate migrated data to reduce errors and ensure integrity.

Utilizing AI to Refactor Code and Streamline Complex Migrations



Codebase Analysis

AI tools analyze existing codebases to identify areas for improvement and optimization effectively.



Automated Refactoring

AI suggests and applies code refactors to improve quality and maintainability with minimal human input.



Migration Streamlining

AI automates complex system migration tasks, reducing manual workload and ensuring smoother transitions.

REDUCING PRESSURE ON TRADITIONAL MIGRATION TOOLS WITH AI CAPABILITIES



Overcoming Scale Limitations

AI augmentation helps traditional migration tools manage large-scale transitions more effectively and efficiently.

Handling Complexity

Integrating AI reduces the complexity of cloud migrations, simplifying processes for organizations.

Enhanced Flexibility and Efficiency

AI-powered tools increase flexibility and operational efficiency during cloud transitions.

Key Take-Aways

- **SaaS:** Quick wins for small workloads
- **PaaS:** Balance, requires refactoring
- **IaaS:** Heavy lifting for large workloads
- **Key:** Match workload to service model

Conclusion: Strategic Cloud Database Migration for Diverse Workloads

Choose the Cloud Model

Select cloud deployment models that best fit specific workload requirements for optimal performance and scalability.

Do Cost Planning

Carefully plan and manage costs to ensure migration remains within budget while meeting performance goals.

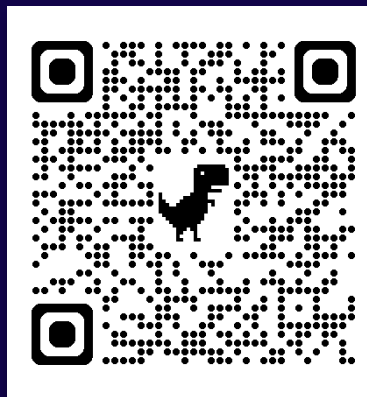
Use Advanced Tools and AI

Leverage AI and advanced migration tools to enhance efficiency and reduce risks during cloud database migration.

Thank you

Kellyn and Rene

Kellyn:



Rene:



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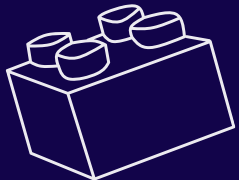
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